Yuanfan Chen

646-963-8426 | yc2942@cornell.edu | github | linkedin

Education

Cornell Tech, Cornell University — New York, NY
Master of Engineering in Computer Science — *Aug. 2025 – May. 2026*

University of Toronto — Toronto, Canada
Honours Bachelor of Science, Computer Science Specialist (GPA: 3.87/4.0) — *Sep. 2020 – May. 2024*

Experience

Machine Learning Engineer — May 2025 – Sep 2025
*Tencent: **Distributed Training and Inference*** — *Beijing, CN*

- Designed **request-level admission control** with **batch-size stepwise profiling** for latency estimation and **SLA-aware priority scheduling**; maintained **100% goodput** under **2–10× QPS bursts** by enforcing per-class latency budgets across mixed-priority traffic.
- Identified **high prefix redundancy** in production traffic and adopted **SGLang with RadixAttention** for prefix caching; implemented **prefill-first chunking** over decode-first, reducing **P99 TTFT by ∼40%** without sacrificing decode throughput.
- Optimized CUDA attention kernels for **GQA** (A100) and **DeepSeek FlashMLA** sparse attention (H200); profiled with **Nsight Compute** to localize memory-hierarchy bottlenecks and redesigned thread-block tiling, shared-memory layout, and warp scheduling; achieved ∼**20% throughput gain** on GQA and ∼**15%** on FlashMLA over baselines.

ML Systems Researcher — Sep 2025 – Present
Cornell University — *New York, NY*

- Quantified that reasoning workloads cause **50% output-length misprediction** in LLM request schedulers; demonstrated via simulation that this misrouting degrades **P99 TTFT by up to 8**×, revealing a fundamental gap in production serving systems.
- Developed **PTA** (Prefill→Think→Answer) three-stage disaggregated serving; identified that each phase impacts UX differently, enabling **differential SLO constraints**; independently configured **DVFS frequency**, **KV-Cache allocation**, and **parallelism** per stage — preserving user-perceivable latency (**TTFAT**) via Answer-phase frequency protection under power caps while improving overall throughput through stage-level resource tuning.
- Built a quantitative **GPU frequency–performance model** by systematically profiling **DVFS** frequency vs. power, throughput, and latency curves on A100/H100; designed an **online optimization framework** that jointly schedules **LC/Flex/BE** request classes, maximizing throughput and QoS under **instantaneous power caps** and **cumulative energy budget** constraints.
- Applied **γ-Boost Queuing Theory** to design a prediction-free, tail-aware scheduler with **KV-cache-aware preemption**; reduced **tail TTFT/TTLT by 37–60%** while improving throughput by **1.01–1.12**× on Azure production traces.

Systems Researcher — Jan 2024 – Feb 2025
Far Data Lab, University of Toronto — *Toronto, CA*

- Co-designed **dpBento**, a modular benchmark framework that standardizes performance evaluation across heterogeneous DPUs (**BlueField-2/3**, **OCTEON TX2**), from micro-level primitives to end-to-end cloud database offloading.
- Profiled **RocksDB** compaction and decompression paths end-to-end; offloaded compute-heavy stages to DPU, reducing **host CPU utilization by ∼40%** and achieving ∼**3**× **decompression throughput** via hardware acceleration.

Technical Skills

Programming: Python, C++, CUDA, Java **AI Infra**: vLLM, SGLang, KV-Cache, PD separation
Distributed: Raft, ClickHouse, HDFS **Systems**: Linux internals, CPU/GPU arch, Nsys/Nsight
Networking: epoll, multiplexing, async I/O, KCP **Databases**: RocksDB, MySQL, Redis

Publications

- **Making Sense of DPU Performance for Cloud Data Processing: [Experiment, Analysis & Benchmark]**. Jiasheng Hu, Chihan Cui, **Yuanfan Chen**, Philip A. Bernstein, Jialin Li, Qizhen Zhang. arXiv:2504.05536 (2025).